

Networking Basics

1. OSI Model Layers & Functions

The OSI (Open Systems Interconnection) Model is a framework used to understand how data travels across a network.

1. Physical Layer – Transmits raw bits through cables, signals, and hardware.
2. Data Link Layer – Handles MAC addresses and communication between devices on the same network.
3. Network Layer – Responsible for IP addressing and routing.
4. Transport Layer – Provides reliable data transfer using TCP and fast communication using UDP.
5. Session Layer – Establishes, manages, and terminates communication sessions.
6. Presentation Layer – Handles data encryption, decryption, and formatting.
7. Application Layer – Provides services such as HTTP, FTP, SMTP, and DNS.

2. TCP/IP Protocol Suite

TCP/IP is the foundation of modern networking and the Internet.

Layers of TCP/IP Model:

1. Network Access Layer
2. Internet Layer
3. Transport Layer
4. Application Layer

TCP (Transmission Control Protocol):

- Reliable communication
- Error checking
- Connection-oriented

UDP (User Datagram Protocol):

- Faster communication
- No connection required
- Used in streaming and gaming

3. DNS (Domain Name System)

DNS translates domain names into IP addresses.

Example:

google.com → 142.250.x.x

Functions:

- Converts website names to IP addresses
- Makes browsing easier for users
- Supports internet communication

4. HTTP and HTTPS

HTTP (HyperText Transfer Protocol):

- Used for web communication
- Data is transmitted in plain text
- Uses Port 80

HTTPS (HyperText Transfer Protocol Secure):

- Secure version of HTTP
- Uses SSL/TLS encryption
- Uses Port 443
- Protects sensitive information

5. IP Addressing

An IP Address uniquely identifies a device on a network.

Example:

192.168.1.10

Types:

- Public IP Address
- Private IP Address

IPv4 Example:

192.168.1.1

IPv6 Example:

2001:db8::1

6. Subnetting

Subnetting is the process of dividing a large network into smaller networks.

Benefits:

- Improved performance
- Better security
- Efficient IP address management

Example:

192.168.1.0/24

7. NAT (Network Address Translation)

NAT allows multiple devices to share a single public IP address.

Advantages:

- Conserves public IP addresses
- Improves security
- Commonly used in home and office routers

Example:

A Wi-Fi router translates private IP addresses to a public IP address for internet access.